myVLA: An Experimental Study of Visual Adaptation and Multimodal Fusion

Behavior cloning on LIBERO-Spatial | English experimental report
Repository evidence audited on 11 November 2026

Abstract. This report consolidates six myVLA configurations evaluated at five training checkpoints on LIBERO-Spatial. The study varies visual-encoder fine-tuning and the depth of bidirectional vision-language fusion while keeping the text encoder frozen. The audited corpus comprises 500 demonstrations, 62,250 training frames, and 30 evaluations of 100 episodes each. At 10,000 training steps, success ranges from 79% to 96%. The six-block model with a trainable visual encoder (V2.2) achieves 96/100 successes, compared with 84/100 for the frozen baseline and 92/100 for visual fine-tuning without fusion. Fusion depth alone does not yield a monotonic improvement. We report checkpoint trajectories, paired comparisons, task-level outcomes, and training-resource observations. Interpretation is limited by one recorded training seed, a small evaluation sample, incomplete historical hardware metadata, and a microbatch difference in V2.1.

1. Experimental objectives and model variants

The project implements a compact language-conditioned action-chunk policy using pretrained CLIP ViT-B/16 encoders [1] and evaluates manipulation behavior in LIBERO [2]. The present experiments are joint multi-task behavior cloning on the Spatial suite; they do not constitute a lifelong-learning or unseen-task evaluation. The analysis asks whether visual adaptation helps, whether deeper fusion improves control, and whether offline training loss predicts closed-loop success.

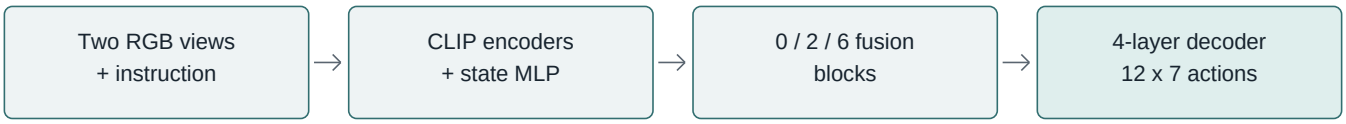


Figure 1. Shared policy structure. The state embedding joins the decoder memory after visual-language fusion. Text is frozen in every variant; visual fine-tuning and fusion depth define the experimental groups. [S3]

Model	Visual encoder	Fusion blocks	Total params.	Trainable params.
V1	Frozen	0	154.24 M	4.62 M
V1.1	Frozen	2	157.40 M	7.78 M
V1.2	Frozen	6	163.72 M	14.10 M
V2	Trainable	0	154.24 M	90.42 M
V2.1	Trainable	2	157.40 M	93.58 M
V2.2	Trainable	6	163.72 M	99.89 M

Table 1. Canonical variants, identified by checkpoint architecture rather than historical folder names. Totals include the original CLIP projection parameters, which are frozen and unused by the policy forward pass. [S2, S3]

The current V4 configuration adds eight fusion blocks, but no completed V4 evaluation is present. MetaWorld MT50 support is implemented, but no MT50 training/evaluation results are available in the audited output directories. Both are excluded from quantitative comparisons.

2. Dataset, implementation, and environment

2.1. Demonstrations and model inputs

All six checkpoint manifests contain the same 500 training trajectories from 10 LIBERO-Spatial tasks, with 50 demonstrations per task and 62,250 frames in total. Every manifest entry belongs to the training split. No held-out demonstration validation set is used. Frames are sampled uniformly with replacement, so longer trajectories contribute more samples than shorter ones. A nominal 10,000-step run with effective batch 64 draws 640,000 frame samples, equivalent to about 10.28 dataset-sized sample counts, rather than 10.28 sequential epochs. [S2, S3]

Each sample provides two RGB views (agent view and wrist view), a task instruction, and an eight-dimensional state consisting of end-effector position, axis-angle orientation, and two gripper positions. The inspected HDF5 observations are 128 x 128 RGB; preprocessing vertically flips LIBERO images, resizes them to 224 x 224 by bicubic interpolation, and applies CLIP channel normalization. State and continuous-action means and standard deviations are computed from training demonstrations. There is no configured image shift or color jitter. [S2-S4]

2.2. Recorded software and current host

Component	Version or configuration	Evidence and scope
Python / package	Python 3.10.20; myvla 0.1.0	Python: current environment; package: evaluation metadata
Learning stack	PyTorch 2.6.0; Transformers 4.57.6; NumPy 1.26.4	Recorded in 24/30 evaluation metadata headers
LIBERO simulation	robosuite 1.4.1; MuJoCo 2.3.7; relative OSC_POSE	Evaluation metadata; LIBERO Git revision is not recorded
CLIP backbone	ViT-B/16; revision 57c21647...e4ae5f3	Same full pinned revision in all checkpoint configs
Current CPU / RAM	Intel Core i9-12900H; about 15.35 GiB host RAM	Observed on report date; historical run host not certified
Current GPU	NVIDIA GeForce RTX 3060 Laptop GPU; 6,144 MiB	Current host only; historical GPU model is not logged
Current OS / CUDA	Ubuntu 24.04.5 LTS; CUDA build 12.4; driver 595.99.02	Current installed PyTorch is 2.6.0+cu124

Table 2. Environment provenance. Installed versions and the current host are distinguished from historical experiment records; cloud-style V2.1 paths do not establish the cloud GPU model. [S1, S2, S4]

2.3. Shared representation and output head

Each camera contributes 196 CLIP patch tokens; CLS tokens are dropped. Features are projected to width 256 and augmented with camera embeddings. Fusion variants use bidirectional cross-attention with zero-initialized residual gates and eight attention heads. The decoder consumes 392 visual tokens plus language and state summaries. Four Transformer decoder layers predict 12 future seven-dimensional actions in parallel from learned action queries. Dropout is 0.1. The six continuous outputs control relative pose; the last output is a gripper logit. [S2, S3]

3. Training and evaluation protocol

Setting	Recorded value
Optimizer / schedule	AdamW; weight decay 0.001; 500-step warmup; cosine decay to 10% of peak LR
Learning rates	Fusion/action head: 1e-4; trainable CLIP visual encoder: 1e-5
Training budget	10,000 nominal optimizer iterations; checkpoint every 2,000 steps
Batching / workers	16 x 4 accumulation for five groups; V2.1 uses 8 x 8; effective batch 64; two workers
Numerics / stabilization	FP16 autocast with FP32 parameters; gradient scaling; gradient clipping at 1.0
EMA / seed	EMA decay 0.999 for the trainable modules; training seed 42
Evaluation / control	10 tasks x 10 episodes; seed 42; 220 control steps; 10 settling steps
Action execution	Execute 12 predicted actions before observing again; reset action queue per episode

Table 3. Checkpoint-derived training settings and recorded evaluation protocol. Effective batch equality does not remove all numerical and sampling differences between microbatch layouts. [S1-S3]

3.1. Masked behavior-cloning objective

For each sampled frame, the target is the next 12 actions. Windows that extend beyond a trajectory end are padded and masked. The loss averages Smooth L1 over the six normalized motion dimensions and adds binary cross-entropy for gripper open/close classification; both terms exclude padded timesteps. With valid-timestep mask m, the implementation computes:

$$L = [\text{sum } m\ (\text{mean}_{j=1..6}\ \text{SmoothL1}(a_j, y_j) + \text{BCE}(g, y_g))] / \text{sum } m.$$

During rollout, continuous outputs are denormalized, the gripper logit is thresholded at zero to obtain -1 or +1, and all actions are clipped to [-1, 1]. Training loss is evaluated on demonstration states, whereas rollout success depends on the full sequence of states induced by the policy. [S3]

3.2. Outcome definition and pairing

A rollout succeeds when the environment success predicate becomes true within the horizon. The reported aggregate is the mean task success rate, which equals episode-level accuracy because every task contributes 10 episodes. All 30 episode logs contain the same task/episode initial-state indices, enabling paired comparisons across models and checkpoints. The 3,000 logged trials reuse 100 evaluation conditions; they are not 3,000 independent test conditions.

3.3. Evidence validation and uncertainty

For every evaluation, progress.json contains exactly 100 Boolean outcomes and agrees with both log.json and the 100 unique episode records. Twenty-four evaluations retain a full metadata header confirming EMA weights and the protocol above. The six older evaluations for V1/V1.1 at 6k, 8k, and 10k lack that header: their outcomes and initial-state pairing are verified, but the historical weight selection and complete protocol cannot be independently re-established from those files. Wilson 95% intervals are descriptive binomial summaries; paired exact McNemar tests are exploratory, unadjusted, and do not account for task clustering or training-seed variability. [S1]

4. Aggregate closed-loop results

Model	2k	4k	6k	8k	10k	Best observed
V1	2%	76%	89%	81%	84%	89% at 6k
V1.1	13%	78%	80%	83%	84%	84% at 10k
V1.2	19%	74%	80%	82%	79%	82% at 8k
V2	26%	91%	89%	87%	92%	92% at 10k
V2.1	18%	70%	88%	86%	85%	88% at 6k
V2.2	26%	86%	95%	91%	96%	96% at 10k

Table 4. Success rates over 100 episodes per checkpoint. Each cell is recomputed from committed episode outcomes. Best observed scores are post hoc maxima on the same evaluation set and are not independent validation estimates. [S1]

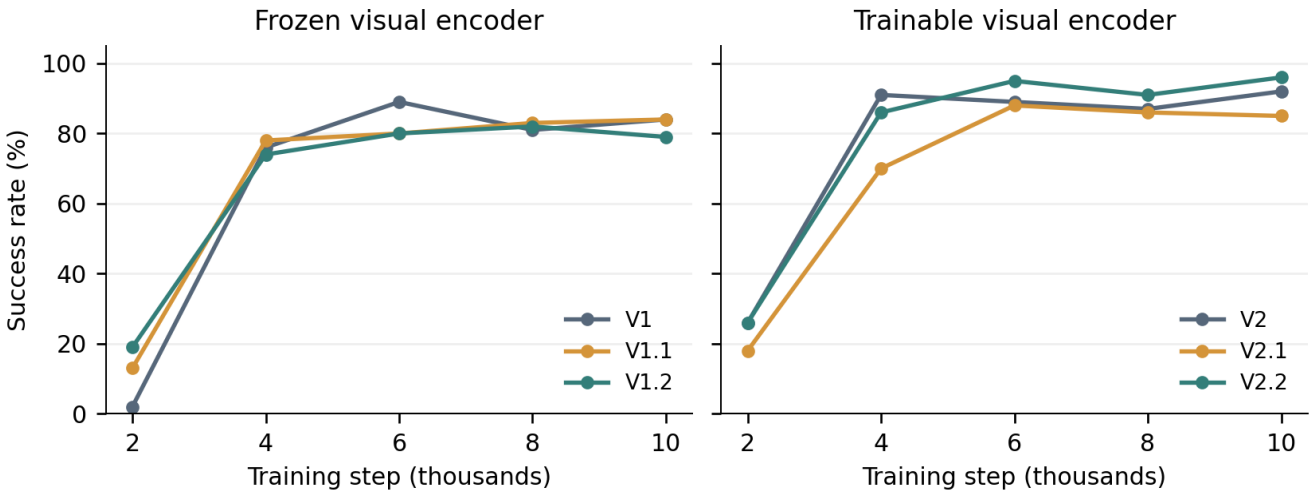


Figure 2. Success trajectories across the five checkpoint stages. Colors indicate 0, 2, and 6 fusion blocks within each visual-training regime. The final checkpoint is used for the main matched-budget comparison. [S1]

4.1. Visual fine-tuning and fusion interact

At the final checkpoint, visual fine-tuning raises success from 84% to 92% without fusion (+8 percentage points, pp), from 84% to 85% with two blocks (+1 pp), and from 79% to 96% with six blocks (+17 pp). Thus, the largest observed gain occurs when a deeper fusion module and visual adaptation are used together. V2.2 also exceeds the no-fusion trainable model V2 by 4 pp and the two-block V2.1 model by 11 pp; these are observed differences in this evaluation, not established population effects.

4.2. Training stage matters

At 2k steps, all variants perform poorly (2%-26%). By 4k, V2 reaches 91%, ahead of V2.2 at 86%. V2.2 becomes the best observed configuration from 6k onward (95%, 91%, 96%). Success is non-monotonic: V1 peaks at 6k and drops by 5 pp by 10k; V2.1 drops by 3 pp from its 6k maximum. These fluctuations caution against treating later checkpoints or lower loss as automatic improvements.

Averaging the three late checkpoints gives 80.3%-84.7% for frozen-vision groups and 86.3%-94.0% for trainable-vision groups. These averages summarize correlated stages of the same training runs; they are not repeated-seed estimates.

5. Ablation comparisons and uncertainty

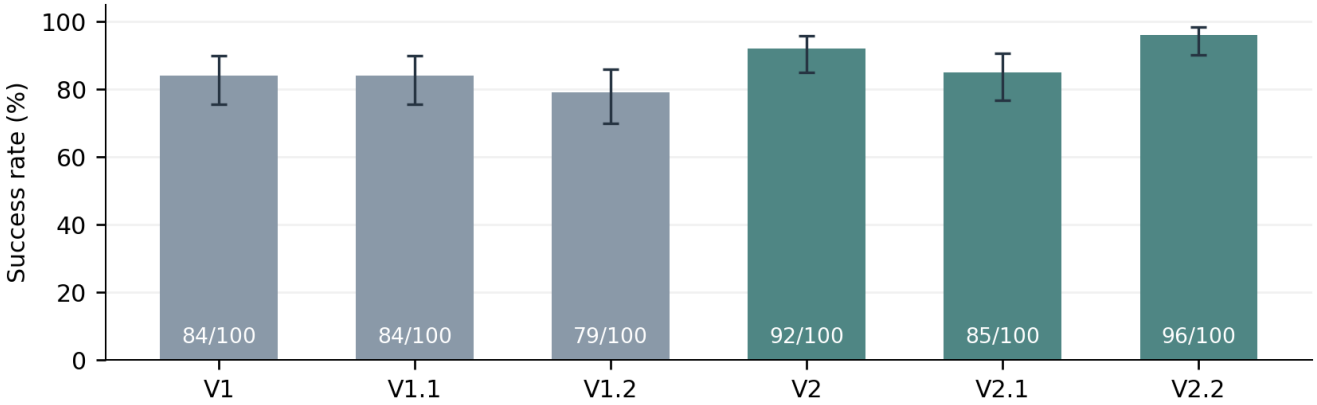


Figure 3. Final-checkpoint success with descriptive 95% Wilson intervals (n=100 per model). The intervals summarize episode outcomes under the recorded evaluation conditions and do not quantify variation across independently trained models. [S1]

Contrast (B minus A)	Gain	B-only wins	A-only wins	Exact p
V2 vs V1	+8 pp	15	7	0.134
V2.1 vs V1.1	+1 pp	13	12	1.000
V2.2 vs V1.2	+17 pp	19	2	0.00022
V2.2 vs V2	+4 pp	7	3	0.344
V2.2 vs V2.1	+11 pp	14	3	0.013

Table 5. Paired final-checkpoint contrasts. A B-only win is an initial condition where B succeeds and A fails. Exact two-sided McNemar p-values use only discordant pairs; they are nominal, not multiplicity-adjusted, and assume independent pairs despite shared task structure. [S1]

5.1. What the evidence supports

The strongest within-record contrast is V2.2 versus V1.2: 19 evaluation conditions improve and two regress, producing +17 pp (nominal p=0.00022). This supports prioritizing joint visual adaptation for this architecture, while remaining conditional on this single training seed and task collection. By contrast, the 4 pp difference between V2.2 and V2 consists of seven wins and three regressions (p=0.344); this sample does not establish a reliable superiority of six fusion blocks over no fusion.

V2.2 obtains a descriptive interval of 90.2%-98.4%, compared with 85.0%-95.9% for V2 and 75.6%-89.9% for V1. The broad overlap and only 10 episodes per task limit fine-grained rankings. The V2.1 comparison is additionally affected by its 8 x 8 batching layout and resumed training history. Equal effective batch sizes should therefore be described as budget matching, not complete experimental control.

5.2. Final checkpoint versus model selection

Using 10k for all groups avoids selecting each model at a different budget, although the report itself remains retrospective. Choosing V1 at its 6k maximum yields 89%, while V2.2 at its best stage yields 96%; the resulting 7 pp gap is smaller than the 12 pp final-checkpoint gap. Neither selection rule substitutes for evaluating a chosen model on fresh seeds and unused initial conditions. A future protocol should predefine the selection rule and reserve a separate final test set.

6. Task-level performance and remaining failures

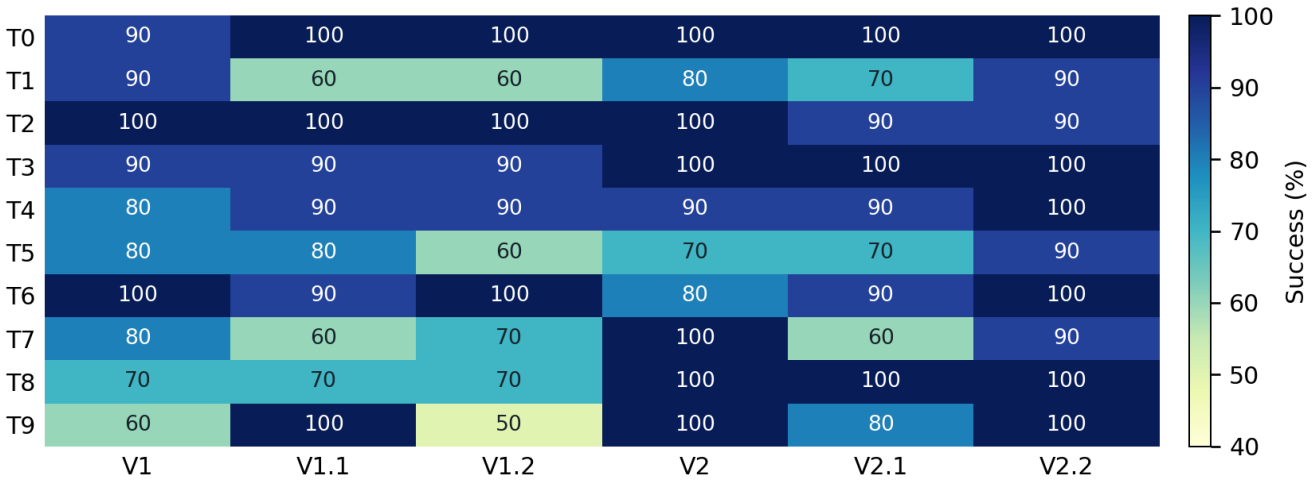


Figure 4. Final-checkpoint success (%) by task. Each cell uses only 10 episodes, so values change in increments of 10 pp. Task IDs follow the common episode logs. [S1]

Task	Initial bowl location	V1	V2	V2.2
T0	Between plate and ramekin	90%	100%	100%
T1	Next to ramekin	90%	80%	90%
T2	Table center	100%	100%	90%
T3	On cookie box	90%	100%	100%
T4	Inside top drawer	80%	90%	100%
T5	On ramekin	80%	70%	90%
T6	Next to cookie box	100%	80%	100%
T7	On stove	80%	100%	90%
T8	Next to plate	70%	100%	100%
T9	On wooden cabinet	60%	100%	100%

Table 6. Task key and selected final-checkpoint results. All tasks instruct the robot to pick up the black bowl and place it on the plate; only the initial spatial relation differs. [S1, S4]

6.1. Where the aggregate improvement occurs

Relative to V1, V2.2 improves the wooden-cabinet task from 60% to 100% and the next-to-plate task from 70% to 100%. It matches V1 on the next-to-ramekin and next-to-cookie-box tasks, and drops from 100% to 90% at table center. Relative to V2, gains are concentrated on the ramekin (+20 pp), next-to-cookie-box (+20 pp), next-to-ramekin (+10 pp), and drawer (+10 pp) tasks, offset by 10 pp drops at table center and on the stove.

V2.2 succeeds in all 10 episodes on six tasks and in nine episodes on the remaining four. Its four failures occur at T1, T2, T5, and T7. The outcome logs do not classify grasp slips, collisions, missed localization, or placement errors. These mechanisms therefore cannot be assigned as observed failure causes without reviewing trajectories or videos.

7. Optimization behavior and resource observations

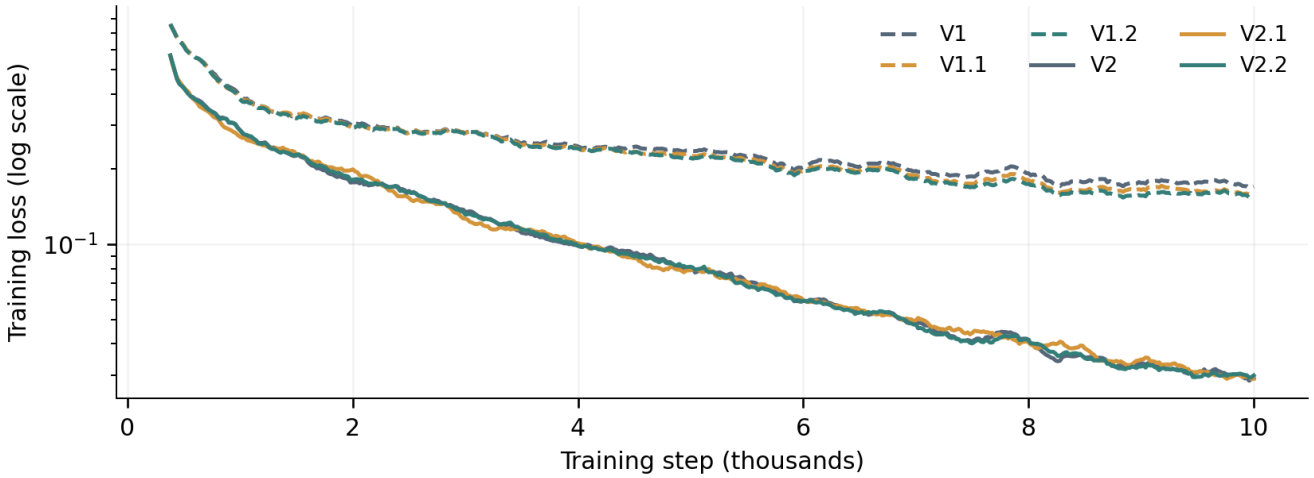


Figure 5. Logged training loss, smoothed with a trailing mean over 20 logged observations (roughly 400 training iterations). Dashed curves freeze the visual encoder; solid curves train it. The loss is computed on raw online parameters rather than the evaluated EMA parameters. [S2, S3]

Model	Tail loss	Logged peak GPU memory	Logged training time
V1	0.1705	0.86 GiB	2.30 h
V1.1	0.1591	0.99 GiB	2.44 h
V1.2	0.1553	1.43 GiB	1.53 h
V2	0.0292	4.87 GiB	3.28 h
V2.1	0.0290	3.65 GiB	4.92 h (sum of 3 sessions)
V2.2	0.0299	5.53 GiB	5.30 h

Table 7. Tail loss is the mean of the final 20 unique logged steps. Memory is the maximum allocated-GPU-memory value reported by the training logs, not total device usage. V2.1 time sums the final elapsed counter of each resumed session; other times use the final counter. These are heterogeneous run observations, not controlled hardware benchmarks. [S2]

7.1. Demonstration fitting versus closed-loop behavior

Visual fine-tuning reduces tail loss to about 0.029-0.030, versus 0.155-0.171 with frozen visual features. Yet V2.1 has the lowest tail loss (0.0290) and only 85% final success, whereas V2.2 has a slightly higher loss (0.0299) and 96% success. Likewise, frozen V1.2 fits demonstrations better than V1 by this loss summary but achieves lower final success (79% versus 84%). The loss ranking therefore fails to reproduce the rollout ranking.

Several differences explain why the two quantities need not align: the loss averages local action errors on demonstration states; rollout success is a thresholded long-horizon outcome on policy-induced states; and evaluation uses EMA where recorded, while the loss comes from raw training weights. These are properties of the objective and evaluation design, rather than proof that any single failure is caused by distribution shift.

7.2. Capacity and cost

Trainable parameters rise from 4.62 M in V1 to 99.89 M in V2.2. The highest logged memory rises from 0.86 GiB to 5.53 GiB. V2.2 records 5.30 h of training versus 3.28 h for V2, but historical GPU identity and system load are not captured. A speedup or cost-efficiency claim cannot be inferred from these times alone.

8. Interpretation, causal hypotheses, and limitations

8.1. Why visual adaptation may help

Observed: trainable-vision variants have substantially lower behavior-cloning loss, and the final success gain over the corresponding frozen model is +8, +1, or +17 pp. **Hypothesis:** action supervision may adapt CLIP patch features to task-relevant geometry, contact cues, and camera appearance. The large V1.2-to-V2.2 gain is consistent with this explanation, but feature probes, localization measurements, and repeated-seed ablations would be needed to demonstrate it. These experiments do not measure representation quality directly.

8.2. Why more fusion is not uniformly better

Observed: with frozen vision, final success is 84%, 84%, and 79% for 0, 2, and 6 fusion blocks. With trainable vision, the corresponding results are 92%, 85%, and 96%. **Hypothesis:** fusion may be most useful when the visual features can co-adapt to the language-conditioned control objective. When vision remains fixed, extra capacity may mainly reduce demonstration error without improving the geometry of action decisions. Optimization sensitivity and finite-sample variation are alternative explanations. The recorded data do not justify a monotonic depth-scaling claim.

8.3. Why early checkpoints are weak and late success fluctuates

Observed: 2k success is 2%-26%, followed by large gains at 4k, while several later trajectories dip. **Hypotheses:** early under-training and EMA lag can contribute to the initial deficit. With fixed decay 0.999, the EMA half-life is approximately 693 updates, and the initial parameter copy retains about 13.5% weight after 2,000 successful updates. This arithmetic suggests lag as a plausible mechanism, but there is no raw-versus-EMA evaluation to isolate it. Later fluctuations could reflect policy sensitivity, EMA dynamics, imperfect demonstration coverage, or small-sample noise; they do not by themselves prove overfitting. [S3]

8.4. Limits on causal and generalization claims

Only one recorded training seed is available per architecture, and the same 100 initial conditions are reused across checkpoints. Task outcomes are clustered, and multiple retrospective comparisons increase selection risk. The six older V1/V1.1 evaluations lack full metadata. V2.1 uses different microbatching and has three logged training sessions; its historical output paths also change. All groups train and evaluate within the same 10-task suite, so the study establishes neither unseen-language generalization nor transfer to other LIBERO suites or MT50. The documented earlier validation-split change has no matched archived control in this corpus; no numerical benefit is attributed to that change here.

8.5. Experiments that would resolve the main uncertainties

First, repeat the 2 x 3 visual-training/fusion design with at least three independent training seeds, identical microbatching, and a predeclared final-checkpoint protocol. Second, evaluate more initial states with task-aware uncertainty estimates. Third, compare raw and EMA weights at 2k and 4k to test the lag hypothesis. Fourth, vary action execution length (for example, 1, 4, 8, and 12 steps) with one fixed trained policy to measure replanning effects. Finally, label failures from videos, then evaluate the eight-block V4 and additional suites as separate studies. These are proposed follow-ups; none is counted as a completed result.

9. Conclusions and evidence provenance

The audited experiments favor adapting the visual encoder and show that fusion depth interacts with that adaptation. V2.2 is the best observed model at the shared 10k budget, reaching 96% on the 100 recorded LIBERO-Spatial episodes. Its gain is 17 pp over the matching frozen six-block model, 12 pp over the frozen no-fusion baseline, and 4 pp over the trainable no-fusion model. The last of these differences is small relative to the available sample and is not established as a robust advantage. Training loss is useful for monitoring optimization, but closed-loop evaluation is necessary to compare control performance.

The report is an analysis of existing artifacts, not a new training or simulation run. Numerical results were recomputed from outcome records; model identity and training settings were read from checkpoint metadata, which take precedence over renamed folders and current default YAML files. No outcome was inferred for an unevaluated model.

9.1. Local evidence map

ID	Repository evidence	Use in this report
S1	outputs/eval/libero_spatial_v*/ckpt{step}/seed42/ progress.json, log.json, episodes.jsonl	30 evaluations; aggregate/task outcomes; paired initial conditions; metadata coverage
S2	outputs/train/libero_spatial_v*/ checkpoint_{step}.pt, manifest.json, metrics.jsonl or log.jsonl	Canonical architecture; actual training settings; demonstration counts; loss/time/memory logs
S3	src/myvla/model.py, data.py, preprocess.py, train.py, evaluate.py, ema.py	Policy structure; sampling; loss; action decoding; rollout; EMA implementation
S4	data/libero_spatial/*_demo.hdf5; pyproject.toml; installed package inventory	Example image schema; dependency specification and current local versions
S5	configs/train_cfg_v4.yaml; README.md; model-version history document	Planned eight-block configuration and contextual documentation; not substituted for raw outcomes

Table 8. Paths are relative to the project root. In S1/S2, v* expands to v1, v1_1, v1_2, v2, v2_1, or v2_2, and step expands to 2000, 4000, 6000, 8000, or 10000. Canonical labels use dots in the report and underscores in file paths.

9.2. Reproducibility details and unresolved records

All models pin CLIP to revision 57c216476eefef5ab752ec549e440a49ae4ae5f3. Archived folders use older labels, including a two-block trainable configuration named v4 in cloud paths; this corresponds to canonical V2.1, not the current eight-block V4. Matching trajectory identities and lengths are verified across the six manifests, but file paths and timestamps alone do not establish a cryptographic hash of every HDF5. Historical GPU/driver details and the LIBERO source Git revision remain unrecorded. The evidence cutoff is 11 November 2026.

References

[1] A. Radford et al. Learning Transferable Visual Models From Natural Language Supervision. *Proceedings of ICML*, PMLR 139:8748-8763, 2021. proceedings.mlr.press/v139/radford21a.html

[2] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. 2023. arXiv:2306.03310. Used for benchmark context; no published baseline scores are compared in this report.